

E-Commerce SQL Data Analysis Project

Project Overview

This project is an E-Commerce Data Analysis project developed using SQL to analyze customer behavior, product performance, orders, sales, revenue, and purchasing patterns. The main objective of this project is to transform raw e-commerce data into meaningful insights using SQL queries.

The project focuses on analyzing different aspects of an e-commerce business, including customer orders, product sales, revenue generation, category performance, customer spending, order value, and purchasing frequency.

The project is divided into **six analytical modules**, where each module focuses on a specific business problem and gradually builds SQL analysis skills.

Through this project, SQL concepts such as **SELECT, WHERE, GROUP BY, ORDER BY, aggregate functions, JOINS, subqueries, CASE statements, date functions, and data analysis techniques** were applied.

Main Objectives:

- Analyze customer purchasing behavior.
- Identify top-performing products and categories.
- Calculate sales and revenue-related metrics.
- Analyze customer spending and order patterns.
- Identify high-value customers and orders.
- Study monthly sales trends and purchasing frequency.

Tools & Technologies:

- SQL / MySQL
- MySQL Workbench
- E-Commerce relational database
- SQL queries for data analysis

Module 1 — Basic E-Commerce Data Analysis

Module 1 focuses on understanding the basic structure of the e-commerce database and performing fundamental data analysis using SQL.

Important Points:

- Understanding customers, products, orders, and order-item data.
- Retrieving and filtering relevant records using SQL.
- Performing basic calculations using aggregate functions.
- Sorting and grouping data to identify useful patterns.

Key SQL Concepts:

SELECT, WHERE, ORDER BY, GROUP BY, aggregate functions.

Module 2 — Product & Sales Analysis

Module 2 focuses on analyzing products and sales-related information to understand product performance.

Important Points:

- Analyzing products based on quantity sold.
- Identifying sales patterns across products.
- Calculating product-level sales metrics.
- Comparing products based on their performance.

Key SQL Concepts:

Aggregate functions, GROUP BY, ORDER BY, filtering, calculations.

Module 3 — Sales & Customer Analysis

Module 3 focuses on combining customer, order, and product information to perform more detailed sales analysis.

Important Points:

- Product-wise quantity sold analysis.
- Sales analysis using table JOINS.
- Customer-wise total spending analysis.
- Connecting orders with order-product information.

Key SQL Concepts:

JOIN, SUM(), GROUP BY, ORDER BY, multi-table analysis.

Module 4 — Customer & Order Analysis

Module 4 focuses on understanding customer purchasing behavior and order-related metrics.

Important Points:

- Customer-wise order count analysis.
- Customer-wise total spending.
- Customer-wise sales analysis.
- Top-selling product analysis.
- Customer-wise Average Order Value (AOV).

Key SQL Concepts:

JOIN, COUNT(), SUM(), AVG(), GROUP BY, subqueries.

Module 5 — Revenue & Sales Trend Analysis

Module 5 focuses on analyzing revenue generation and identifying sales trends.

Important Points:

- Product-wise revenue analysis.
- Category-wise sales analysis.
- Top-selling product analysis.
- Customer-wise revenue analysis.
- Monthly sales trend analysis.

Key SQL Concepts:

Revenue calculations, `SUM()`, `GROUP BY`, date functions, `ORDER BY`, `JOINS`.

Module 6 — Advanced Customer & Order Analysis

Module 6 focuses on advanced analysis of customer purchasing patterns and order values.

Important Points:

- Customer segmentation based on spending power.
- Customer purchase frequency analysis.
- Category-wise Average Order Value.
- High-value order analysis.
- Identifying meaningful customer and order patterns.

Key SQL Concepts:

`CASE`, aggregate functions, subqueries, `HAVING`, `AVG()`, `COUNT()`, advanced filtering.

SQL Skills & Key Learnings

SQL Skills Developed

During this project, I developed practical knowledge of SQL for data analysis and business problem-solving.

Key SQL Skills:

- Data retrieval using `SELECT`.
- Data filtering using `WHERE` and `HAVING`.
- Data aggregation using `SUM()`, `COUNT()`, and `AVG()`.
- Grouping data using `GROUP BY`.
- Sorting results using `ORDER BY`.
- Combining multiple tables using `INNER JOIN` and other `JOIN` operations.
- Using subqueries for advanced analysis.
- Applying conditional logic using `CASE`.
- Performing date-based analysis for sales trends.
- Calculating revenue, spending, order value, and other business metrics.

Key Learnings

This project helped me understand how SQL can be used to solve real-world business problems using structured data.

Major Learnings:

- Learned how to analyze relational e-commerce data.
- Improved understanding of database relationships.
- Learned how to connect multiple tables using `JOINS`.
- Developed the ability to convert business questions into SQL queries.
- Learned to analyze customer purchasing behavior.
- Gained practical experience in sales and revenue analysis.
- Improved problem-solving and analytical thinking.
- Learned how to extract meaningful business insights from raw data.

Project Outcome

By completing this six-module project, I gained hands-on experience in using SQL for **data cleaning, data exploration, customer analysis, product analysis, sales analysis, revenue analysis, and business insights generation.**

Overall, this project strengthened my foundation in SQL and prepared me for further learning in Data Analytics.

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